

Josue Tapia Hernández

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EDUCATION

Tecnológico de Monterrey <i>BS in Mathematics and Data Science</i> — GPA: 97/100 — Academic Scholarship (Top 5%). Coursework: Data Structures & Algorithms, Statistical Inference, Optimization, Deep Learning, Topology.	Guadalajara, Mexico <i>Expected June 2027</i>
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EXPERIENCE

Co-Founder & CTO <i>JATA — AgTech Startup</i> — Took an AgTech product from hackathon win to paying customers: built and shipped a Dockerized FastAPI backend (7+ routes: auth, ingest, sensors, predictions, admin) on Railway with JWT auth and Postgres row-level security via Supabase — in production for 2 paying commercial greenhouse clients across 5 growing seasons , now onboarding more. — Built a real-time telemetry pipeline: Telegraf ingest service streaming 16 sensor features into the API, surfaced in a React dashboard polling every 30s — operators track sensor health, anomalies, and forecast-vs-actual yield without engineering support. — Shipped the ML layer behind the API: CNN-RNN yield forecaster (3 residual conv blocks + 3-layer LSTM) reaching $\mathbf{R^2 = 0.70}$, $\mathbf{MAPE = 10.7\%}$ over 365 weekly production records ; wrote pytest integration tests covering the ingest, auth, and prediction endpoints.	Sep 2025 – Present <i>Guadalajara, Mexico</i>
Undergraduate Machine Learning Researcher <i>Tecnológico de Monterrey — Advisor: Prof. Andrés Ramírez Morales</i> — Built an $SU(2)$/Spin(3)-equivariant transformer from scratch over the $Cl(3,0)$ geometric algebra for boosted top-quark jet tagging — a 2.5K-line PyTorch library (equivariant linear, attention, LayerNorm, and geometric-bilinear MLP primitives; GATr blocks) covered by a 60+ test pytest suite including explicit equivariance checks. — First author on “ <i>$SU(2)$ Equivariance for Top Jet Classification</i> ” (in preparation, targeting Phys. Rev. D) — best variant reaches test AUC 0.9811 ± 0.0009 at 189K parameters , matching our own L-GATr rerun (0.9813) on the same benchmark with 5.7× fewer parameters ; AUC 0.9850 / 93.8% accuracy on the full 1.2M-jet reference benchmark. — Designed and ran an 8-variant ablation over Hydra-configured experiment branches — authored the precompute pipelines deriving subset-axis bivectors, Casimir invariant masses, and Cambridge–Aachen splitting-plane phases from raw 4-momenta, injected into otherwise-unused multivector grades.	Dec 2025 – Present <i>Guadalajara, Mexico</i>

PROJECTS

Hive — Autonomous MLOps Agent Platform [GitHub] <i>Python, FastAPI, React/TypeScript, LangGraph, Docker</i> 2026 — Designed a 13-agent LangGraph orchestrator (17-node StateGraph with sequential, parallel, and loop phases plus a conflict resolver) that turns a plain-English prompt plus a dataset into a full training run: EDA, architecture selection, GPU provisioning on Vast.ai, Docker/CUDA setup, LLM-generated training script, teardown. — Shipped the full stack: 7-route FastAPI backend streaming over WebSockets into a React + TypeScript dashboard (live pipeline graph, GPU gauges, loss charts, cost KPIs), SSH-executed jobs across 9+ regions , and a 25-test pytest suite — cutting a manual MLOps workflow from ~2 hours to under 10 minutes and GPU spend by 99% (\$0.029/hr spot vs. \$3.50/hr on-demand).	
Berries TDA Forecasting [GitHub] <i>Python, giotto-tda, PyTorch, React/TS, GitHub Actions, Vercel, AWS S3</i> 2026 — Built a self-updating forecasting product on the FRED Berry PPI series (2008–2026): a GitHub Actions cron pulls each new monthly observation, refits the model, and redeploys a React/TypeScript dashboard on Vercel — refits archived to S3 via GitHub OIDC (no long-lived AWS keys). — Research core: Takens delay embedding ($d=12, \tau=3$) + Vietoris–Rips persistent homology (H_0, H_1) generating 157 topological features as exogenous inputs to LSTM and multi-task models — TDA features beat the pure-LSTM baseline on MAE; H_1 persistence peaks over 2008–2012, quantifying crisis-era nonlinear cyclicity. Shipped with Vitest unit tests and a public model leaderboard.	
Process-Verified RL for Lean 4 Theorem Proving [GitHub] <i>PyTorch, PEFT/LoRA, LeanDojo, Lean 4, React</i> 2026 — Implemented a GRPO training loop from scratch (PPO-clipped loss, KL penalty, reference logits via adapter disabling) fine-tuning DeepSeek-Prover-V1.5-RL with LoRA on miniF2F-lean4, with LeanDojo executing tactics against a live Lean 4 kernel for ground-truth reward. — Designed a novel cross-generation reward graph: rollouts become nodes linked by Jaccard similarity over normalized tactics, propagating credit from successful to failed proofs so a correct tactic is rewarded even inside a failed attempt — parameter-free, no extra Lean calls. Shipped with a React front end and end-to-end tests.	

AWARDS & HONORS

1st Place, UST Hackathon — winning entry became JATA, now a revenue-generating AgTech company	2025
ICPC National Regional Contest — Honourable Mention (2×) — Top 17% nationally	2024, 2025
National Olympiad in Informatics (OMI) — Bronze Medal — Top 15% — selected for IOI national training	Nov 2022
National Physics Olympiad (ONF) — Bronze Medal — Top 20% — selected for IPhO national training	Nov 2022
National Math Olympiad (OMM) — Honourable Mention (2×) — solution officially cited by UADY	2021, 2022

SKILLS

Languages: Python, TypeScript/JavaScript, C++, SQL, R, Bash, Lean 4.
Backend & Frontend: FastAPI, REST API design, WebSockets, JWT auth, React, Tailwind, Vite, PostgreSQL/Supabase (row-level security).
Infra & Tooling: Docker, Git/GitHub, GitHub Actions (CI/CD), Vercel, Railway, AWS S3 + OIDC, Vast.ai, Hydra, Linux/Unix, pytest, Vitest.
AI & ML: PyTorch, TensorFlow, scikit-learn, PEFT/LoRA, LangGraph, DSPy, Gemini & Claude APIs, XGBoost, YOLOv8, giotto-tda, LeanDojo, Clifford/geometric algebra. Spoken: Spanish (Native), English (C1).